LINEAR ALGEBRA & CALCULUS

(23A54101)

UNIT - II

Eigenvalues, Eigenvectors & Orthogonal Transformation

Topics covered:

- Eigenvalues and eigenvectors and their properties
- Diagonalization of a matrix
- Cayley-Hamilton theorem (without proof)
- Inverse and powers of a matrix by Cayley-Hamilton theorem
- Quadratic forms and nature of quadratic forms
- Reduction of a quadratic form to canonical form by orthogonal transformation

Handwritten Notes - B.Tech (JNTUA R23)

1. Eigenvalues and Eigenvectors

For a square matrix A (order n), a non-zero vector X is an EIGENVECTOR and the scalar λ the corresponding EIGENVALUE if $AX = \lambda X$, i.e. $(A - \lambda I)X = 0$. Non-zero X exists iff $|A - \lambda I| = 0$:

$$|A - \lambda I| = 0 \text{ (characteristic equation)}$$

For a 3 x 3 matrix it expands to

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0$$

S_1 = trace = $a_{11} + a_{22} + a_{33}$; S_2 = sum of the principal minors of order 2 (= sum of the co-factors of the diagonal elements); $|A|$ = determinant. For 2 x 2: $\lambda^2 - S_1\lambda + |A| = 0$.

Method: (1) form and solve the characteristic equation for λ; (2) for each λ solve $(A - \lambda I)X = 0$ by row reduction – the free variables give the eigenvector(s).

Properties of eigenvalues:

- Sum of eigenvalues = trace of A; product of eigenvalues = $|A|$.
- A and A^T have the same eigenvalues. A singular $\Leftrightarrow$ 0 is an eigenvalue.
- Triangular / diagonal matrix: eigenvalues are the diagonal elements.
- If λ is an eigenvalue of A then: $K\lambda$ for KA; λ^m for A^m; $1/\lambda$ for A^{-1}; $|A|/\lambda$ for adj A; $\lambda + K$ for $A + KI$; $p(\lambda)$ for $p(A)$.
- Eigenvectors belonging to DISTINCT eigenvalues are linearly independent.
- Real symmetric matrix: eigenvalues are real and eigenvectors of distinct eigenvalues are ORTHOGONAL.
- Eigenvalues of a skew-symmetric matrix are zero or purely imaginary; of an orthogonal matrix have modulus 1.

Problem 1: find the eigenvalues and eigenvectors of $A = [\,4\ 1\,;\,2\ 3\,]$.

$S_1 = 7$, $|A| = 10$: $\lambda^2 - 7\lambda + 10 = 0 \rightarrow (\lambda - 2)(\lambda - 5) = 0$. $\lambda = 2, 5$.

$$\lambda = 2: A - \lambda I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 1/2 \\ 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad (\lambda = 2)$$

$$\lambda = 5: A - \lambda I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (\lambda = 5)$$

Check: sum = 7 = trace ✓, product = 10 = $|A|$ ✓.

Problem 2: find the eigenvalues and eigenvectors of $A = [\,8\ -6\ 2\ ;\ -6\ 7\ -4\ ;\ 2\ -4\ 3\,]$.

$S_1 = 18$, $S_2 = (56 - 36) + (24 - 4) + (21 - 16) = 45$, $|A| = 0$.

$$\lambda^3 - 18\lambda^2 + 45\lambda = 0 \rightarrow \lambda(\lambda - 3)(\lambda - 15) = 0$$

Eigenvalues $\lambda = 0, 3, 15$.

$$\lambda = 0: A - \lambda I = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad (\lambda = 0)$$

$$\lambda = 3 : A - \lambda I = \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \quad (\lambda = 3)$$

$$\lambda = 15 : A - \lambda I = \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \quad (\lambda = 15)$$

The eigenvectors are mutually orthogonal (A is symmetric) – e.g. $(1,2,2)\cdot(2,1,-2) = 2 + 2 - 4 = 0$ ✓.

Problem 3 (repeated eigenvalue): find the eigenvalues and eigenvectors of $A = [\,2\ 2\ 1\,;\,1\ 3\ 1\,;\,1\ 2\ 2\,]$.

$S_1 = 7$, $S_2 = 4 + 3 + 4 = 11$ … (cofactors of diagonal), $|A| = 5$.

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0 \rightarrow (\lambda - 1)^2(\lambda - 5) = 0$$

$\lambda = 1$ (repeated twice) and $\lambda = 5$.

$$\lambda = 1 : A - \lambda I = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$A - I$ has rank 1, so TWO free variables ($x_2 = s$, $x_3 = t$): $x_1 = -2s - t$.

$$X = s\begin{bmatrix}2\\-1\\0\end{bmatrix} + t\begin{bmatrix}1\\0\\-1\end{bmatrix} \quad (\lambda = 1)$$

$$\lambda = 5 : A - \lambda I = \begin{bmatrix}-3 & 2 & 1\\ 1 & -2 & 1\\ 1 & 2 & -3\end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix}1 & 0 & -1\\ 0 & 1 & -1\\ 0 & 0 & 0\end{bmatrix}$$

$$X = k\begin{bmatrix}1\\1\\1\end{bmatrix} \quad (\lambda = 5)$$

For the repeated root we obtained two independent eigenvectors, so this A is diagonalizable.

Problem 4: eigenvalues of $A = [\,1\ 1\,;\,0\ 1\,]$. Triangular $\Rightarrow \lambda = 1, 1$.

$$\lambda = 1 : A - \lambda I = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$$

Only ONE independent eigenvector $(1, 0)^T$ for the double root $\Rightarrow$ this matrix is NOT diagonalizable (defective).

Problem 5 (using properties): the eigenvalues of A (3×3) are 1, 2, 3. Find the eigenvalues of (i) $A^2 + 2A + I$ (ii) A^{-1} (iii) adj A and $|A|$.

- (i) $p(\lambda) = \lambda^2 + 2\lambda + 1 = (\lambda + 1)^2 \rightarrow 4, 9, 16$.
- (ii) $1/\lambda \rightarrow 1, 1/2, 1/3$.
- (iii) $|A| = 1 \cdot 2 \cdot 3 = 6$; $|A|/\lambda \rightarrow 6, 3, 2$ (so $|\text{adj } A| = 36$).

2. Diagonalization of a Matrix

A is DIAGONALIZABLE if there is a non-singular P with $P^{-1}AP = D$ (diagonal). P is the MODAL MATRIX whose columns are n linearly independent eigenvectors; D has the eigenvalues in the same order on its diagonal.

- A is diagonalizable ⇔ it has n linearly independent eigenvectors (always true if all eigenvalues are distinct, or if A is symmetric).
- Then $A = PDP^{-1}$ and $A^m = PD^mP^{-1}$ with $D^m = \text{diag}(\lambda_1^m, \lambda_2^m, \ldots)$.
- Real symmetric A can be diagonalised by an ORTHOGONAL matrix: $P^TAP = D$.

Problem 6: diagonalize $A = [\,1\ 1\ 3\,;\,1\ 5\ 1\,;\,3\ 1\ 1\,]$.

$$\lambda^3 - 7\lambda^2 + 36 = 0 \rightarrow (\lambda + 2)(\lambda - 3)(\lambda - 6) = 0$$

Eigenvectors: $\lambda = -2 \rightarrow (1, 0, -1)^T$; $\lambda = 3 \rightarrow (1, -1, 1)^T$; $\lambda = 6 \rightarrow (1, 2, 1)^T$ (found as in Problem 1–3).

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 2 \\ -1 & 1 & 1 \end{bmatrix}, \quad P^{-1}AP = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

So $P^{-1}AP = \text{diag}(-2, 3, 6) = D$.

Problem 7: diagonalize $A = [\,4\ 1\,;\,2\ 3\,]$ and hence find A^5.

$$P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \quad P^{-1} = \begin{bmatrix} 1/3 & -1/3 \\ 2/3 & 1/3 \end{bmatrix}$$

$$A^5 = PD^5P^{-1} = \begin{bmatrix} 2094 & 1031 \\ 2062 & 1063 \end{bmatrix}$$

Problem 8: diagonalize $A = [\,2\ 2\ 1\,;\,1\ 3\ 1\,;\,1\ 2\ 2\,]$ (repeated root).

$$P = \begin{bmatrix} -2 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, P^{-1}AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Columns of P: two eigenvectors of $\lambda = 1$ and one of $\lambda = 5$ – three independent vectors ⇒ diagonalizable.

3. Cayley–Hamilton Theorem (without proof)

Every square matrix satisfies its own characteristic equation. If $\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$ is the characteristic equation of A then

$$\boxed{A^3 - S_1A^2 + S_2A - |A|I = 0}$$

Uses: (i) verify the theorem; (ii) find A^{-1}; (iii) find higher powers A^m and polynomials in A (reduce using the equation).

Problem 9 (verification): verify the theorem for $A = [1\ 2; 2\ 1]$.

$\lambda^2 - 2\lambda - 3 = 0$ ($S_1 = 2$, $|A| = -3$). To show $A^2 - 2A - 3I = 0$:

$$A^2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}, 2A = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}, 3I = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$A^2 - 2A - 3I = [5\ 4; 4\ 5] - [2\ 4; 4\ 2] - [3\ 0; 0\ 3] = [0\ 0; 0\ 0]$ ✓ Theorem verified.

Problem 10 (inverse): find A^{-1} using the theorem, for $A = [2\ 1\ 1; 0\ 1\ 0; 1\ 1\ 2]$.

$S_1 = 5$, $S_2 = (2 - 0) + (4 - 1) + (2 - 0) = 7$, $|A| = 3$.

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0 \Rightarrow A^3 - 5A^2 + 7A - 3I = 0$$

Multiply by A^{-1}: $A^2 - 5A + 7I - 3A^{-1} = 0 \Rightarrow$

$$\boxed{A^{-1} = \frac{1}{3}(A^2 - 5A + 7I)}$$

$$A_2 = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} \quad A^{-1} = \tfrac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ 0 & 1 & 0 \\ -1/3 & -1/3 & 2/3 \end{bmatrix}$$

Problem 11 (polynomial in A): for the same A evaluate $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.

Divide the polynomial by $\lambda^3 - 5\lambda^2 + 7\lambda - 3$:

$$= A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + A^2 + A + I$$

The brackets are zero by Cayley–Hamilton, so the value is $A^2 + A + I$:

$$A^2 + A + I = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

Problem 12 (powers): for $A = [1\ 2; 3\ 4]$ find A^4 using Cayley–Hamilton.

$\lambda^2 - 5\lambda - 2 = 0 \Rightarrow A^2 = 5A + 2I$. Then $A^3 = 5A^2 + 2A = 27A + 10I$; $A^4 = 27A^2 + 10A = 145A + 54I$.

$$A^4 = 145A + 54I = \begin{bmatrix} 199 & 290 \\ 435 & 634 \end{bmatrix}$$

(Check by direct multiplication $A^2 \cdot A^2$ ✓.)

4. Quadratic Forms and their Nature

A QUADRATIC FORM in n variables is a homogeneous polynomial of degree 2: $Q = X^T A X$ with A SYMMETRIC. Diagonal elements of A = coefficients of x_i^2; off-diagonal $a_{ij} = a_{ji} = \frac{1}{2}$ (coefficient of $x_i x_j$).

Problem 13: write the matrix of $Q = 3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$.

$$Q = X^T A X, \quad A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

Canonical form: by a linear transformation $X = PY$, Q becomes a sum of squares $Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_r y_r^2$. Rank r = number of non-zero terms; index p = number of positive terms; signature = $2p - r$.

Nature	Eigenvalues of A	Principal minors
Positive definite	all $\lambda > 0$	all leading minors > 0
Negative definite	all $\lambda < 0$	minors alternate $-, +, -, \dots$
Positive semi-definite	all $\lambda \geq 0$, at least one $= 0$	≥ 0
Negative semi-definite	all $\lambda \leq 0$, at least one $= 0$	—
Indefinite	both positive and negative λ	otherwise

Problem 14: find the nature of $Q = x^2 + 2y^2 + 3z^2 + 2xy - 2yz$ by leading principal minors.

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

$$\Delta_1 = 1,\ \Delta_2 = 1,\ \Delta_3 = 2$$

All leading minors are positive ⇒ Q is POSITIVE DEFINITE.

Problem 15: nature of $Q = 2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx$ using eigenvalues.

Characteristic equation $\lambda^3 - 6\lambda^2 + 9\lambda = 0 \rightarrow \lambda(\lambda - 3)^2 = 0$: eigenvalues 0, 3, 3.

All $\lambda \geq 0$ and one is 0 ⇒ POSITIVE SEMI-DEFINITE ; canonical form $3y_2^2 + 3y_3^2$, rank 2, index 2, signature 2.

Problem 16: nature of $Q = x^2 + 4xy + y^2$.

$A = [1\ 2\ ;\ 2\ 1]$: $\lambda^2 - 2\lambda - 3 = 0 \rightarrow \lambda = 3, -1$. One positive, one negative ⇒ INDEFINITE. Canonical form $3y_1^2 - y_2^2$: rank 2, index 1, signature 0.

5. Reduction to Canonical Form by Orthogonal Transformation

Procedure for $Q = X^T AX$ (A real symmetric):

1. Find the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of A.
2. Find the eigenvectors; for DISTINCT λ they are already orthogonal – for a repeated λ choose orthogonal vectors (Gram–Schmidt).
3. Normalise each vector (divide by its length) and use them as the columns of the ORTHOGONAL matrix P ($P^TP = I$).
4. Put $X = PY$. Then $Q = Y^TDY = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$ (canonical form), with $P^TAP = D$.

Problem 17: reduce $Q = 6x^2 + 3y^2 - 4xy$ to canonical form by an orthogonal transformation.

$A = [\,6\ -2\ ;\ -2\ 3\,]$: $\lambda^2 - 9\lambda + 14 = 0 \rightarrow \lambda = 2, 7$.

$\lambda = 2$: $4x - 2y = 0 \rightarrow X_1 = (1, 2)^T$; $\lambda = 7$: $-x - 2y = 0 \rightarrow X_2 = (2, -1)^T$. ($X_1 \cdot X_2 = 0$ ✓). Normalising by $\sqrt{5}$:

$$P = \frac{1}{\sqrt{5}}\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix},\ P^T A P = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\boxed{X = PY : Q = 2y_1^2 + 7y_2^2}$$

Nature: positive definite (both eigenvalues positive), rank 2, index 2, signature 2.

Problem 18: reduce $Q = 3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ to canonical form by an orthogonal transformation.

Characteristic equation: $\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0 \rightarrow (\lambda - 2)(\lambda - 3)(\lambda - 6) = 0$.

$$\lambda = 2 : X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\quad \lambda = 3 : X_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\quad \lambda = 6 : X_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$X_1 \cdot X_2 = 0$, $X_2 \cdot X_3 = 0$, $X_1 \cdot X_3 = 0$ – mutually orthogonal. Lengths $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$.

$$P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix},\ P^T A P = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$\boxed{X = PY : Q = 2y_1^2 + 3y_2^2 + 6y_3^2}$$

Nature: POSITIVE DEFINITE; rank 3, index 3, signature 3.

Problem 19 (repeated eigenvalue): reduce $Q = 2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx$ by an orthogonal transformation.

$\lambda = 0, 3, 3$. $\lambda = 0 : X_1 = (1, 1, 1)^T$. $\lambda = 3 : -x - y - z = 0$ (rank 1). Choose two ORTHOGONAL solutions: $X_2 = (1, -1, 0)^T$ and $X_3 = (1, 1, -2)^T$ ($X_2 \cdot X_3 = 0$ ✓, both $\perp X_1$).

$$P = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{bmatrix}, P^T AP = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$X = PY : Q = 0 \cdot y_1^2 + 3y_2^2 + 3y_3^2$$

Positive semi-definite ; rank 2, index 2, signature 2.

Problem 20: reduce $Q = x^2 + 3y^2 + 3z^2 - 2yz$ to canonical form.

A = diag-block : eigenvalues 1, 2, 4. $X_1 = (1,0,0)^T$ ($\lambda = 1$) ; $\lambda = 2$: $y + \ldots \rightarrow$ $X_2 = (0, 1, 1)^T$; $\lambda = 4$: $X_3 = (0, 1, -1)^T$. Normalised:

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$X = PY : Q = y_1^2 + 2y_2^2 + 4y_3^2 \text{ (positive definite, rank 3, index 3)}$$

Important Questions – Unit II

1. Find eigenvalues and eigenvectors of a 2 x 2 / 3 x 3 matrix (distinct and repeated roots).
2. Properties of eigenvalues; find eigenvalues of A^2, A^{-1}, adj A, $A + kI$.
3. Diagonalize a matrix and find a power A^n.
4. State Cayley-Hamilton theorem; verify it; find A^{-1} and powers.
5. Evaluate a matrix polynomial using Cayley-Hamilton.
6. Find the matrix of a quadratic form; find its nature, rank, index and signature.
7. Reduce a quadratic form to canonical form by an orthogonal transformation.